

Title,Authors,Year,DOI or URL,Abstract,Source

"Object-centred automated compliance checking: a novel, bottom-up approach", "Omar Doukari; David Greenwood; Kay Rogage; Mohamad Kassem", 2022, 10.36680/j.itcon.2022.017, "Building Information Modelling (BIM) is changing how built assets are delivered and operated. A built asset is represented as a set of objects, each with an identity, attributes, and relations. This object-oriented nature enables new approaches for ensuring compliance with a range of requirements (e.g. industry guidelines, project-specific requirements, building codes and standards). Furthermore, bottom-up design approaches are known to be more suitable for quality control and error detection. Based on an adapted simulated annealing concept, this paper proposes an automated compliance checking classification and identifies a set of desired characteristics that such methods should fulfill. It then demonstrates a bottom-up object-centered approach for automated model checking with a corresponding plugin prototype. The approach enables four key processes and satisfies all desired characteristics of compliance checking methods, including content validation, model completeness, smart object checking, and design option checking. Two case studies are used to demonstrate feasibility and accuracy: all four steps were successfully completed, and results show savings of around 125 minutes per object compared to traditional manual methods ¹ ² ", "Journal of Information Technology in Construction (ITcon) ³ "

"Multi-ontology fusion and rule development to facilitate automated code compliance checking using BIM and rule-based reasoning", "Jiang Liu; Jianyong Shi; Chaoyu Wang", 2022, 10.1016/j.aei.2021.101449, "This study presents a novel approach that combines multiple ontologies and rule-based reasoning to automate building code compliance checking using BIM. By fusing ontologies from different domains and formalizing compliance rules, the method enhances automated rule interpretation and reasoning in compliance checking. The approach uses BIM models (in IFC format) and a set of fused ontologies to represent regulatory knowledge, enabling an automated reasoning engine to check the BIM model against code requirements. A case study demonstrates that the multi-ontology and rule-based framework can improve the automation and accuracy of compliance checking compared to traditional methods ⁴ ⁵ ", "Advanced Engineering Informatics 51 (Elsevier) ⁴ "

"Automated code compliance checking through building information modelling", "Bilal Aslam; Tariq Umar", 2023, "https://doi.org/10.1680/jstbu.22.00214", "The design, construction, and handover of building projects must comply with local authority rules. Manual assessment of drawings against complex rule sets is cumbersome, inconsistent, costly, and error-prone, causing major delays. This research develops an automated code compliance checking process using BIM. Shortcomings of manual code compliance were identified via a review of existing systems and confirmed through expert interviews. A new automated compliance checking system with BIM integration was then developed and evaluated. The automated BIM-based code checking process reduced review time from about one week to roughly six hours. However, feedback indicated that architects would require higher fees to

produce complete BIM models. Implementing automated code compliance via BIM in municipal processes can raise industry innovation and support progress toward several UN Sustainable Development Goals ⁶ ⁷ ", "Proceedings of the ICE - Structures and Buildings, vol. 177 (2023) ⁸ "

"Automated code compliance checking research based on BIM and knowledge graph", "Junlong Peng; Xiangjun Liu", 2023, 10.1038/s41598-023-34342-1, "Automated code compliance checking plays an important role in advancing the construction industry. While traditional drawing reviews rely on expert identification, this study aims to realize automatic code compliance review by using BIM technology combined with knowledge graph techniques. The proposed method uses a knowledge graph to transform textual code provisions into a computer-recognizable structured format via natural language processing, creating a human-machine-readable knowledge graph representation of the rules. For checking BIM models, the BIM data are exported and represented in RDF (using Protégé), and then mapping rules and review rules are applied to generate a compliance report. An example is presented to validate the feasibility of the approach. The results show that this method effectively addresses the manual dependence and inefficiency of traditional review processes ⁹ ¹⁰ ", "Scientific Reports (Nature) 13:7065 (2023) ¹¹ ¹² "

"Semantic web-based automated compliance checking with integration of Finite Element analysis", "Panagiotis Patlakas; Ioannis P. Christovasilis; Lorenzo Riparbelli; Franco K. T. Cheung; Ermal Vakaj", 2024, 10.1016/j.aei.2024.102448, "Automatic compliance checking (ACC) is a promising approach to ensure building designs meet regulations, especially using Building Information Modeling (BIM) and Industry Foundation Classes (IFC). However, engineering analyses like Finite Element Analysis (FEA) are often used by engineers and have seen little integration into ACC. This work proposes a holistic Semantic Web-based ACC framework that integrates FEA results into the IFC schema. A top-level Building Compliance Ontology (BCO) is introduced to encompass all aspects of regulatory compliance, with a novel focus on modeling regulatory statements (clauses) rather than just building components. To evaluate the approach, a real-world use case is tested: checking a modular steel structure's design for compliance with structural code requirements. An open-source FEA solver is used and its results are embedded into the BIM/IFC model via an open-source middleware. The BCO is extended to cover structural engineering concepts, and appropriate ACC classes and SHACL rules are implemented. Compliance checking is then automatically performed by executing SHACL rules in a Semantic Web environment, successfully verifying the structure's compliance with code provisions ¹³ ¹⁴ ", "Advanced Engineering Informatics 61 (Elsevier) ¹⁵ ¹⁶ "

"Automated Compliance Checking System for Structural Design Codes in a BIM Environment", "Wonhui Goh; Jaeguk Jang; Sang I. Park; Bong-Hyuck Choi; Heuiseok Lim; Goangseup Zi", 2024, 10.1007/s12205-024-1121-5, "This study proposes a framework for developing an automated compliance checking (ACC) system to verify structural design calculations in an openBIM (IFC) environment. The framework comprises four main components: (1) classifying design code provisions by structural component and limit state; (2) preparing a structural analysis model in accordance with the Industry Foundation Classes (IFC) schema; (3) interpreting the code requirements into a machine-readable rule format; and (4) designing the ACC system architecture, focusing

on the flow of rule execution for structural code checks. The framework is demonstrated with a detailed example using ifcOWL, SWRL rules, and a Drools inference engine. A prototype ACC system was developed to check the strength and serviceability requirements of a prestressed concrete girder bridge. The system successfully verified an actual bridge design example. The results indicate that the proposed framework can facilitate the development of ACC systems in structural engineering and can be adapted to various design codes and data sources ¹⁷ ¹⁸ ", "KSCE Journal of Civil Engineering 28:4175-4189 (2024) ¹⁹ ²⁰ "

"Automated Building Information Modeling Compliance Check through a Large Language Model Combined with Deep Learning and Ontology", "Nanjiang Chen; Xuhui Lin; Hai Jiang; Yi An", 2024, 10.3390/buildings14071983, "Ensuring compliance with complex building standards and regulations during design is a significant challenge in the BIM domain. Manual compliance checking is inefficient and error-prone. Recent natural language processing (NLP) and deep learning methods have improved rule interpretation and checking, but they demand extensive feature engineering, large labeled datasets, and high computational resources. This study introduces an innovative automated compliance checking framework that integrates a large language model (LLM) with deep learning and ontology-based knowledge models. The LLM (with few-shot learning capability) is leveraged to reduce the need for large annotated datasets, while deep learning is used to preliminarily classify regulatory text sections, which improves the accuracy of structured information extraction by the LLM. By combining deep learning for text preprocessing with LLM-driven rule interpretation and an ontology for domain knowledge, the framework automates the processing of regulatory texts and significantly reduces manual intervention. In experiments, this approach markedly improves the efficiency and accuracy of compliance checks, offering a scalable and adaptable solution for automated compliance checking in construction ²¹ ²² ", "Buildings (MDPI), 14(7):1983 (2024) ²³ ²⁴ "

"Automatic compliance checking of BIM models against quality standards based on ontology technology", "Zhiliang Ma; Honggang Zhu; Xinglei Xiang; Žiga Turk; Robert Klinc", 2024, 10.1016/j.autcon.2024.105656, "This paper presents an ontology-based approach for automatically checking BIM models against established quality standards. Ontology technology is used to formalize the content of BIM quality standards, enabling computer-interpretable representations of the rules. An inference mechanism is then employed to assess a BIM model's quality compliance with the given standard. A prototype implementation is developed to validate the effectiveness of the proposed framework. In a case study, the system automatically identified compliance issues by reasoning over the ontological representations of both the BIM model data and the quality standard rules ²⁵ ²⁶ ", "Automation in Construction 166 (2024) ²⁷ "

"A Semantic Web and IFC-Based Framework for Automated BIM Compliance Checking", "Lu Jia; Maokang Chen; Chen Chen; Yanfeng Jin", 2025, 10.3390/buildings15152633, "Manual inspection of building design deliverables is time-consuming, labor-intensive, and prone to errors. With the development of BIM technology, automated rule compliance checking has become a trending solution. This paper presents a method that combines Semantic Web technologies with IFC-based data to enhance human-machine collaborative code compliance checking. First, a five-step process (integrated with domain

specifications) is designed to construct a building object ontology that covers most architectural objects in the AEC domain. Next, a set of mapping rules is defined (based on how IFC entities express information) to create a semantic link between the ifcOWL representation of BIM data and the building object ontology. Then, by analyzing building regulations, generic SPARQL query templates are developed for major types of constraints. Finally, a case study using the Jena semantic framework validates the feasibility of the proposed method, demonstrating that the Semantic Web/IFC-based approach can successfully check a BIM model for compliance with regulatory requirements ²⁸

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"BIM-Based Model Checking: A Scientometric Analysis and Critical Review", "Sihao Li; Ziyang Jiang; Zhao Xu", 2025, 10.3390/app15010049, "Building Information Modeling (BIM) has been widely applied throughout project lifecycles in the AEC industry, and errors in BIM models can lead to significant losses. This has driven increasing attention to BIM-based model checking (BMC) technologies. Despite numerous publications describing BMC methods, there has been a lack of systematic analysis and visualization of existing research. This paper conducts a combined scientometric analysis and critical review of BMC-related studies published from 2008 to 2024 in Web of Science and Scopus. The scientometric analysis objectively presents the state and evolution of the field using quantitative data (e.g. publication counts, author networks, citation analysis). The critical review then qualitatively examines the content of the literature, summarizing key research topics and current challenges. Finally, the study identifies promising future research directions for advancing BIM-based model checking. This work provides a comprehensive overview of the development of automated compliance checking (ACC) and related model checking research in BIM, helping researchers understand trends and gaps in the domain ³² ³³ ", "Applied Sciences (MDPI) 15(1):49 (2025) ³⁴ ³² "

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